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Credit institutions are well positioned to handle losses resulting from a carbon tax on agriculture

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In November 2024, a broad majority of the parties in the Danish Parliament reached a political agreement on a national carbon tax on agriculture. The tax, which will be phased in from 2030, will make climate-impacting agricultural production more expensive and thus create an economic incentive for transition and reducing emissions. This analysis examines the impact of the upcoming carbon tax for Danish banks and mortgage credit institutions, which jointly hold loans totalling kr. 274 billion to the agricultural sector. The analysis shows even without lower emissions among farmers as the tax is introduced the institutions are generally well positioned to handle potential losses on their agricultural lending.

Without reductions, agriculture is estimated to account for almost half of Denmark's carbon emissions by 2030

Today, agriculture accounts for a third of Denmark's greenhouse gas emissions (CO_2 equivalent). Without reductions, this share is expected to grow to almost half by 2030. The tax, which will be phased in from 2030 to 2035, will support a green transition in the agricultural sector and thereby reduce emissions over time.

Danish credit institutions are well prepared to handle losses resulting from a carbon tax on agriculture

The financial situation in agriculture has improved since 2016 and debt has decreased significantly. Banks and mortgage credit institutions are also generally well positioned to cover potential losses on loans. This is especially due to the high collateral in underlying assets and their ongoing profits.

Potential losses on agricultural lending can make some institutions more vulnerable to other risks

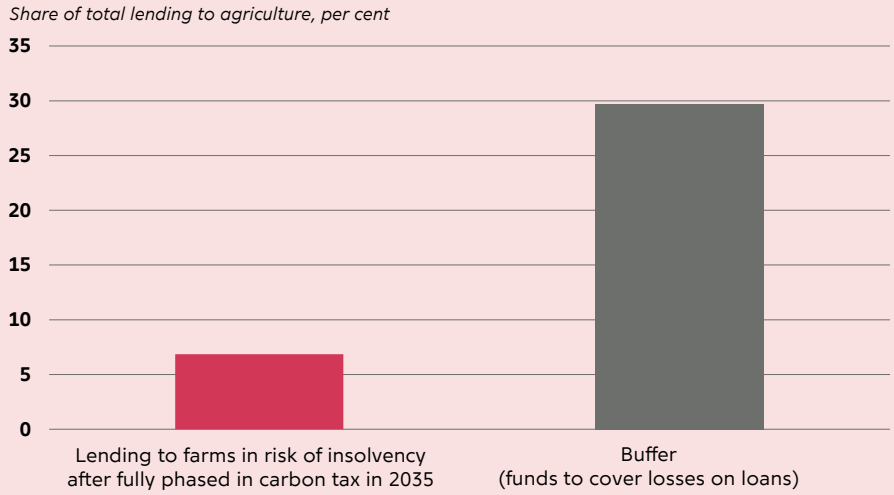
For some banks the losses on loans to agriculture as a result of the carbon tax, can increase the vulnerability to a simultaneous or subsequent economic downturn. However, the announcement and gradual phasing in of the tax will allow institutions to absorb potential losses over several years.

Why is it important?

A green transition of society is important for the fulfilment of Denmark's climate goals in 2030 and beyond. If agriculture does not reduce its emissions, it will be more expensive for the rest of the society to meet its climate goals. With loans totalling kr. 274 billion, upsets in the agricultural sector can potentially affect banks, mortgage credit institutions and financial stability.

Climate change and the green transition of society can generally challenge Danmarks Nationalbank's objectives of stable prices and financial stability in Denmark. This is why it is important for Danmarks Nationalbank to analyse how and to what extent climate change and different outcomes of the green transition may affect the economy and the financial system.

Main chart
Credit institutions are deemed to be able to handle potential losses on agricultural lending



Note: Lending to farms in risk of insolvency in 2035 due to the implementation of the carbon tax from the Green Tripartite Agreement. Lending (before impairment charges) is calculated as of the end of 2023 and projected to 2035 based on developments in farms' financial accounts. The buffers held by institutions to cover loan losses illustrated in the chart consists of already recognised impairment charges, collateral/pledges against underlying assets and annual profit (before tax) in a single financial year (an average of the institutions' financial statements for 2021-23). Both lending and buffers are calculated as a weighted average across institutions.

Source: SEGES Innovation P/S, Expert Group for a Green Tax Reform, Danmarks Nationalbank and own calculations.



Keywords

Climate | Financial stability and financial risks | Banking and mortgage credit | Loans | Households and corporations

01

Reductions in agricultural emissions are key to fulfil Denmark's climate targets

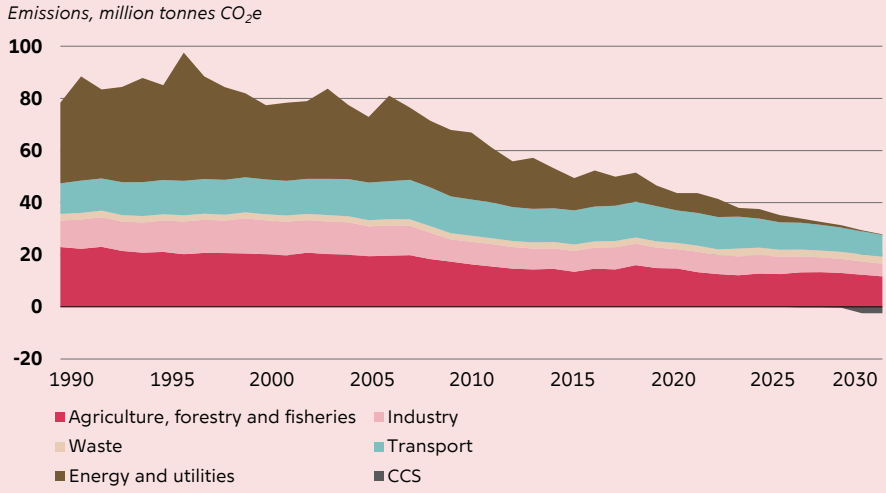
Reducing greenhouse gas emissions is crucial for Denmark to meet its national climate targets. In 2020, the Danish Parliament passed the Danish Climate Act, which sets a target to reduce greenhouse gas emissions by 70 per cent by 2030 compared to 1990 levels. Denmark must also be climate neutral by 2050.[1]

Without reductions, agriculture is expected to account for almost half of Denmark's total greenhouse gas emissions by 2030

The agriculture, forestry and fisheries sector currently accounts for a third of Denmark's total CO_2e emissions. Without reductions, this share is expected to grow to around 46 per cent by 2030, see chart 1.[2] A lack of reductions in agriculture will require further reductions elsewhere to ensure the fulfilment of Denmark's climate targets. Calculations by the Danish Economic Councils in 2020 showed that achieving the 70 per cent reduction target could be about three times more expensive for society if agriculture were exempted from a tax.[3]

CHART 1

Without reductions, agriculture is expected to account for almost half of Denmark's total greenhouse gas emissions by 2030



Note: Projection of Denmark's CO_2e emissions and uptake towards 2030. CCS (carbon capture and storage) are technologies to capture the CO_2 and permanently storing it underground.
Source: Ministry of Climate, Energy and Utilities, *Climate Status and Outlook 2024.*

[1] See the Ministry of Climate, Energy and Utilities for more on Denmark's climate policy and targets (*link*).
[2] See Ministry of Climate, Energy and Utilities, Climate Status and Outlook 2024 (*link* – only in Danish).
[3] Danish Economic Councils, Economy and Environment 2020 – Summary and recommendations (*link*).

Agricultural emissions mainly consist of methane (52 per cent), nitrous oxide (31 per cent) and carbon dioxide (17 per cent). While emissions of carbon dioxide and nitrous oxide in agriculture have decreased by 42 and 33 per cent respectively since 1990, emissions of methane, which originates from the digestive system of livestock, have remained largely unchanged.[4]

The Green Tripartite Agreement includes a carbon tax on agricultural emissions from 2030 onwards

The Agreement on Implementing a Green Denmark was adopted by a broad majority of the Danish Parliament in November 2024. The agreement includes a tax on agricultural emissions of greenhouse gases (CO_2 equivalent[5]) from livestock digestion and manure management. The tax will be introduced in 2030 at a marginal rate of kr. 300 per tonne of CO_2e, gradually increasing to kr. 750 in 2035. This will be combined with a base deduction covering 60 per cent of the average emissions for the given type of animal. This means that the effective tax rate will be kr. 120 in 2030 and kr. 300 in 2035.[6] The base deduction is intended to limit the increase in costs because of the tax and maintain an incentive for individual farms to convert production and use the most climate-efficient operating methods and technologies.

The agreement allocates around kr. 43 billion to the long-term reorganisation of Denmark's land area. The funds will ensure, among other things, that 140,000 hectares of carbon-rich lowland soils are taken out of agricultural production, the creation of 250,000 hectares of new woodland and support for other land conversion (including wetlands and extensification) and 100,000 hectares of virgin woodland. In addition, the agreement sets targets for reductions in nitrogen emissions to ensure good ecological status in Danish coastal waters and inlets.[7] This analysis deals only on the effect of the carbon tax and does not consider the other elements of the political agreement.

With the agreement, Denmark is estimated to reach the 70 per cent target by 2030, but there is still uncertainty about the impact of some policy measures

The agreement is expected to enable Denmark to meet the national 70 per cent reduction target by 2030.[8] However, unexpected challenges in implementing already agreed climate policies or other risks, can lead to the necessary reductions not being achieved. This would require additional climate action to continue to meet climate targets, making the transition more expensive or accelerating it. Danmarks Nationalbank has previously analysed the risks of the green transition and is currently assessing the economic and financial consequences.[9]

The purpose of a carbon tax on agricultural emissions is to create an economic incentive to change production and thereby reduce emissions. A uniform tax across emissions will in general help ensure a market-based incentive to reduce emissions where it is most efficient. This will make the transition as cheap as possible for society.

[4] See Statistics Denmark, Greenhouse gas accounts for agriculture and horticulture, 2022, table DRIVHUS2. Methane emissions come from the digestive system of livestock and associated manure management. Nitrous oxide comes mainly from nitrogenous fertilisers, while carbon dioxide comes from energy consumption and liming of agricultural land. See the Danish Agricultural Agency's page on greenhouse gases from agriculture (*link* – only available in Danish).
[5] Individual greenhouse gases contribute differently to the greenhouse effect depending on their concentration and ability to absorb heat radiation from the Earth. They are therefore converted into a common unit called carbon dioxide equivalents (CO_2e). See more on Statistics Denmark's climate theme page (*link* – only available in Danish) and the Danish Meteorological Institute (DMI) (*link* – only in Danish).
[6] See Agreement on a Green Denmark, June 2024 (*link* – only available in Danish).
[7] See Agreement on Implementing a Green Denmark, Nov. 2024 (*link* – only available in Danish).
[8] The agreement estimates a reduction potential of 1.8-2.6 million tonnes of CO_2e by 2030. See Agreement on Implementing a Green Denmark, Nov. 2024 (*link* – only available in Danish).
[9] See Deis-Bach, Ingholt, Oksbjerg and Jensen, New methodology for assessing economic and financial risks of the green transition, Danmarks Nationalbank Memo, no. 11, November 2024 (*link*).

02

Danish agriculture is highly indebted, but its finances have improved in recent years

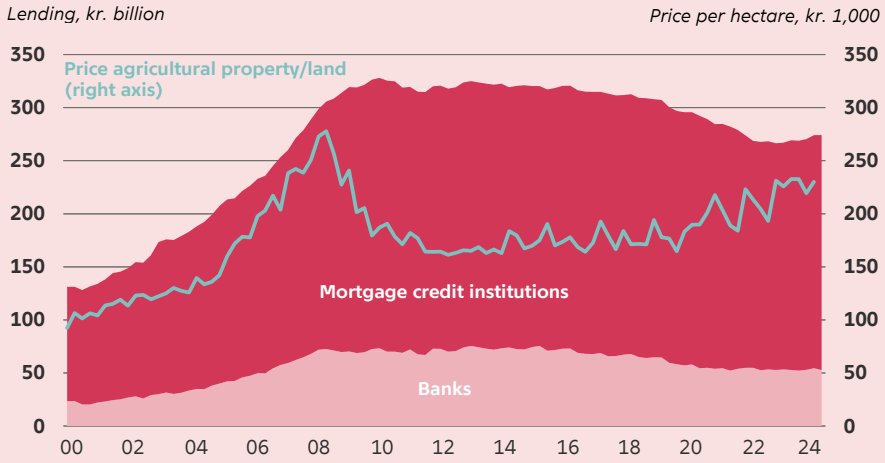
Credit institutions currently lend kr. 274 billion to the agricultural sector, equivalent to 14 per cent of total commercial lending. Mortgage credit institutions account for around 80 per cent of lending to agriculture, see chart 2.

Throughout the 2000s, agricultural debt increased significantly, partly spurred on by rising land and property prices.[10] However, the trend has reversed and lending to agriculture has fallen by kr. 43 billion since 2016, equivalent to almost 14 per cent. Inflation has also contributed to eroding the real value of the debt.

Almost 40 per cent of agricultural lending is to arable farming, while cattle and pig producers each account for around 20 per cent. The rest of the lending is to farms in other livestock farming (11 per cent), forestry (4 per cent), agricultural services (4 per cent) and fisheries (3 per cent).[11]

CHART 2

Lending to agriculture has decreased by 14 per cent since 2016



Note: Total lending to agriculture, forestry and fisheries sector by Danish banks and mortgage credit institutions. Property prices are given by purchase price per hectare in general free trade for agriculture throughout the country. The latest observations are November 2024 for loans and Q3 2024 for agricultural property/land prices.
Source: Danmarks Nationalbank and Statistics Denmark (DataBank, table EJEN77)

[10] See Danmarks Nationalbank, Lave renter og rigelig udlånskapacitet sætter lånestandarder under pres, *Danmarks Nationalbank Analysis (Financial Stability)*, November 2018 (*link* – only available in Danish).
[11] For around 1 per cent of the loans, it is not possible to identify the farm's line of business. All sectors included in the analysis except fishing and hunting.

Farm finances have improved since 2016

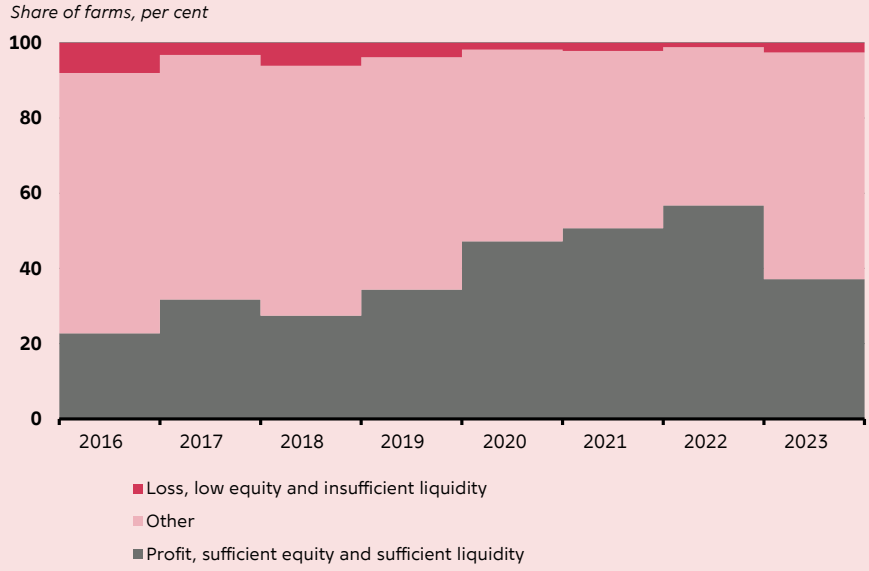
For many farms, the economic situation has improved since 2016. There has been a general consolidation in the industry, and a large proportion of farms have increased their profit.[12]

The number of farms making a profit combined with having sufficient equity and liquidity has grown from 23 per cent in 2016 to 37 per cent in 2023, see chart 3. This trend is largely driven by improved earnings.

Currently, around 2 per cent of Danish farms are estimated to be in risk of insolvency. This is defined by making a loss and having low equity and low liquidity. Being in risk of insolvency does not mean the farm will actually go bankrupt. But it is a strong indication that its finances are under pressure and that the current business model may be unsustainable. See box 1 for a more detailed description of the individual key figures and criteria for when a farm is deemed to be bankrupt.

CHART 3

2 per cent of Danish farms are currently assessed to be at risk of bankruptcy



Note: A farm is characterised as in risk of insolvency if the annual result after financial costs is negative ("loss"), equity amounts to less than 15 per cent of the farm's total assets ("low equity") and liquid assets less depreciation amounts to less than 5 per cent of gross profit ("low liquidity"). The starting point for the analysis is an average of farm accounts for 2021, 2022 and 2023. See box 1 for a more detailed description of the accounting ratios. "Other" in the chart refers to the sum of the six other combinations of the three ratios.
Source: SEGES Innovation P/S and own calculations.

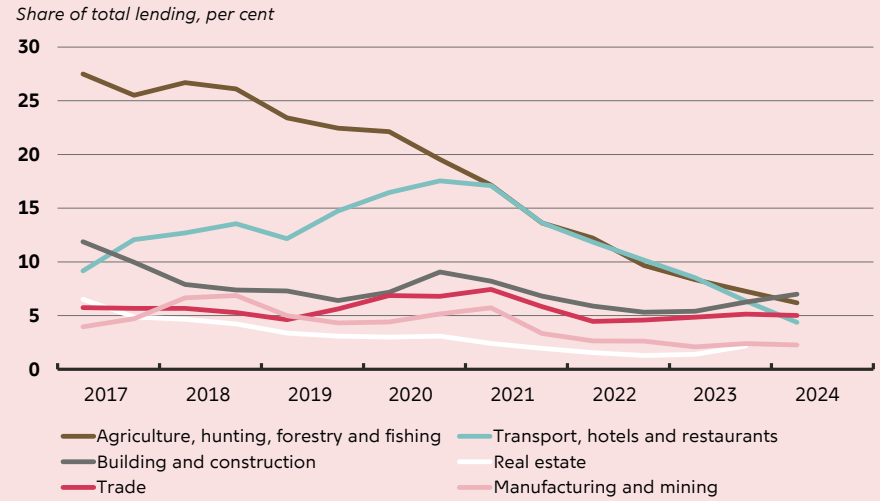
[12] The number of farms in Denmark has decreased from around 42,000 (2010) to around 29,000 (2023), while the total cultivated area has remained almost unchanged over the same period. See Statistics Denmark, DataBank, table BDF11.

The proportion of non-performing agricultural customers has decreased significantly since 2017 and is now on a par with other industries

The proportion of agricultural customers that banks and mortgage credit institutions deem to be non-performing[13] has decreased significantly in recent years. From 2017 to 2024, it decreased from 27 to 6 per cent, which is on a par with business customers in other industries, see chart 4.

CHART 4

The proportion of non-performing agricultural customers has decreased significantly in recent years



Note: Non-performing loans for banks by industry sector. A loan is classified as non-performing if the borrower has not paid interest, fees and/or instalments for more than 90 days or if it is deemed unlikely that the borrower can fully meet their payment obligations. Last observation is from the first half of 2024.
Source: Reports to the Danish Financial Supervisory Authority.

Use of variable-rate loans increases vulnerability to higher interest rates

A sustainable economy is important for an industry like agriculture, which is exposed to many different risks. These include climate-related risks such as weather conditions (e.g. drought) and the introduction of a carbon tax, but also animal diseases and fluctuations in both input prices (e.g. energy) and output prices (e.g. meat and milk). In addition, the interest rates have a major impact on a farm's finances. Most agricultural debt – 90 per cent at the end of 2023 – is at variable rates, increasing vulnerability to higher interest rates. And around half is interest-only loans.

During 2022 and 2023, the average short-term mortgage rate increased significantly (from -0.3 per cent to 4.1 per cent), before falling slightly again in 2024. To the extent that farms have not hedged their interest rate risk, the higher interest rates will have a significant impact on their financial costs and thus their debt servicing capacity. Interest rate rises have contributed to more farms having a deficit in 2023 compared to the previous year. Higher interest rates can increase the proportion of agricultural customers that banks and mortgage credit institutions deem to be non-performing in the future.

[13] A loan is classified as non-performing if the borrower has not paid interest, fees and/or instalments for more than 90 days or if it is deemed unlikely that the borrower can fully meet their payment obligations.

BOX 1

Data on farm finances and definition of insolvent farms

Population
The sample of data on Danish farms was collected by SEGES Innovation P/S and is based on annual data on 4-5,000 Danish farms between 2016-2023. The sample changes from year to year. Lending to the sampled farms accounts for approximately one third of total lending to the industry. Using weights from Statistics Denmark, results in the sample can be calculated for agriculture as a whole. The starting point for the analysis is farm accounts for 2021-23.

Definition of farms in risk of insolvency
The financial robustness of farms is categorised in this analysis based on three accounting ratios. A farm is classified as in risk of insolvency if the following three conditions are met:

1. Negative annual result after financial expenses ("profit")
2. Equity-to-total-assets ratio ("solvency ratio") below 15 per cent
3. Liquidity as a percentage of gross dividend ("liquidity ratio") below 5 per cent

The first two criteria are based on The Danish financial Supervisory Authority's credit rating classification, according to which corporate customers are categorised based on credit quality. The classification states that the credit rating of agricultural customers is based on the farm's ability to generate sufficient earnings from operations, which can be used to pay off debt, and on the real equity and thus the value of the agricultural property.[1]

Being in risk of insolvency does not mean that the farm will actually go bankrupt. But it is a strong indication that the farm is under financial pressure and that the current business model may be unsustainable.

Profit for the year after financial expenses
A carbon tax must be covered by the company's profit. In this analysis, the result for the year is used after financial expenses, but before owner remuneration and tax. This is based on the current structure and financing of the farm. A negative result (loss) indicates that the farm is struggling to generate sufficient revenue.

Equity
A farm's equity will typically act as a buffer against any operational losses. Sufficient equity will also give the farm better opportunities to make necessary investments. In this analysis, the equity ratio is used, which is the amount of equity in relation to total assets. According to the Danish Financial Supervisory Authority's credit rating classification, a solvency ratio of 15 per cent for agriculture is assumed to be an indication of inadequate capital and thus insufficient financial resilience.

Liquidity
A farm's liquidity is important for its ability to fulfil its short-term obligations. Liquidity in this analysis is calculated as profit after tax adjusted for a number of accounting value changes less depreciation and amortisation. Liquidity is an expression of the funds that each farm has for capital investments and debt repayments, for example. The liquidity measure here is calculated relative to the farm's gross profit (revenue). A negative or only slightly positive cash flow (up to 5 per cent of gross profit) is assumed to indicate liquidity pressure.

Use of expanded bankruptcy criterion compared to that of the Expert Group for a Green Tax Reform
This analysis uses an expanded criterion for when a farm is characterised as in risk of insolvency compared to the criterion used by the Expert Group for a Green Tax Reform.[2] This analysis is therefore based on three accounting ratios, whereas the Expert Group used debt ratio and liquidity, but not the result for the year (profit/loss). The debt ratio relates to the ratio between the farm's equity and debt and is therefore an alternative to the equity ratio. The liquidity measure used by the expert group is the same as the one used in this analysis.

[1] See The Danish Financial Supervisory Authority, Guidelines for financial reporting for credit institutions and investment firms etc., version 1 June 2023 (*link* – only available in Danish).
[2] See Expert Group for a Green Tax Reform – Final report, Feb. 2024, section 7.15 (*link* – only available in Danish).

03

The carbon tax will affect the finances of farms to varying degrees

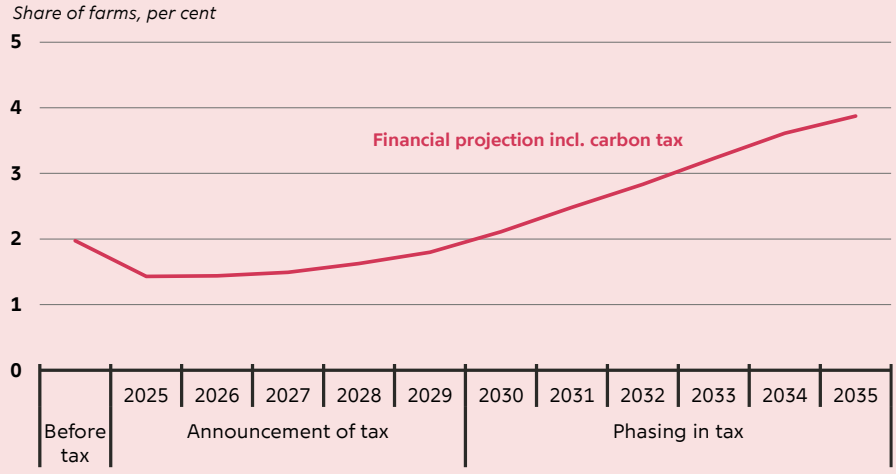
The direct impact of a tax will vary greatly from farm to farm. Emissions are highly dependent on the type of farm, and there are significant differences in the finances of farms.

The calculations of the effect of the carbon tax on agriculture are based on a projection of accounts for a sample of Danish farms. The projection period starts in 2025 when the tax is announced and ends in 2035 when the tax is fully phased in. Emissions from farms subject to the tax cannot be measured directly, but can be calculated based on each farm's activities and the number of livestock animals.

Without lower emissions among farmers as the tax are introduced, the carbon tax, in isolation, will lead to more insolvent farms. In that case, the proportion of insolvent farms is estimated to grow from around 2 per cent in 2025 to almost 4 per cent in 2035, see chart 5. See box 2 for further descriptions of assumptions.

CHART 5

Introduction of the tax can lead to more farms in risk of insolvency



Note: Development in the proportion of farms characterised as in risk of insolvency with the introduction of the carbon tax from the Green Tripartite Agreement. See box 2 for a more detailed description of data and assumptions.
Source: SEGES Innovation P/S, Expert Group for a Green Tax Reform and own calculations

The indirect consequences of a tax may impact the farms' finances

The introduction of a tax is expected to have several-second round effects that will affect farm finances. The extent of the different effects is based on assumptions in GreenREFORM used by the Expert Group for a Green Tax Reform. For more details, see box 2.

A change in the value of land, buildings and machinery are assumed when the tax is announced. Land accounts for more than half of the farmers' total assets, while buildings and machinery accounts for 10 and 7 percent, respectively. Land prices are expected to increase by around 10 per cent at the time of the announcement. This could be seen in the context of an increased demand for land because of national support for afforestation.[14] The increase in land prices explains the decreasing proportion of farms in risk of insolvency in 2025, see chart 5. Land prices are then expected to remain relatively unchanged until 2035. Targets for nitrogen emission reductions, which are politically agreed, can also have an impact on land prices, but this is not included in the analysis.

Land price changes are assumed to be passed on to all farms, while the value of buildings and machinery, which are expected to decrease gradually throughout the period, depend on the farm type. Based on the assumptions in the GreenREFORM model, a tax is also expected to affect both production volumes and prices of agricultural products. A decrease in production would lead to higher prices, and thus, part of the immediate tax burden can partially be passed on to consumers in the form of higher prices. Taken together, these effects will reduce overall earnings in agriculture.

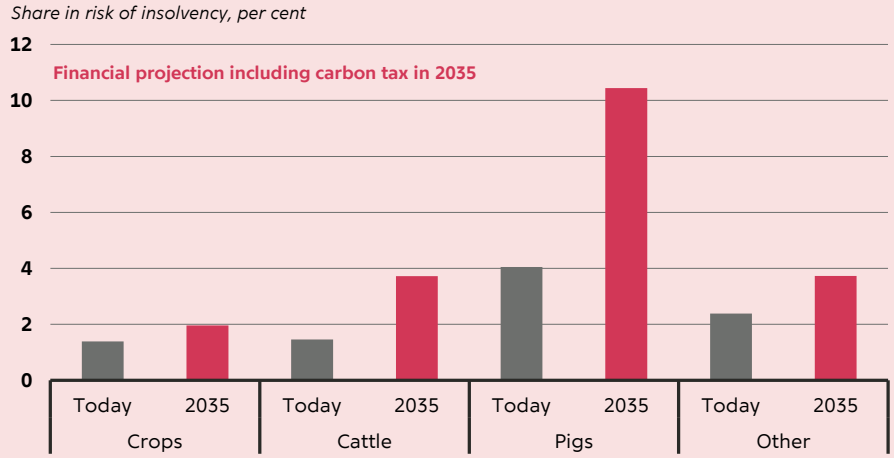
A carbon tax will particularly affect livestock farming

The emissions of farms, and therefore the size of the tax, are highly dependent on the type of operation. It is estimated that livestock farmers in particular will be financially affected by a tax, see chart 6, because these farms typically have higher emissions than farms with plant-based production. In this analysis the effect of a tax is higher for pig farms compared to cattle farms. The development reflects the fact, that cattle producers had relatively good financial results for 2022, which helps improve the starting point for this farm type.

[14] In the political agreement, Agreement on Implementing a Green Denmark, the parties agreed on a national subsidy scheme for private afforestation with a fixed subsidy rate of kr. 75,500/ha. In the political agreement from Nov. 2024, its says that the parties should reach an agreement in the first half of 2025.

CHART 6

Cattle and pig producers have the highest emissions and will therefore be most affected by a tax



Note: Proportion of farms characterised as in risk of insolvency by farm type. The "Other" category includes poultry producers. See box 2 for assumptions behind the calculations.
Source: SEGES Innovation P/S, Expert Group for a Green Tax Reform, Danmarks Nationalbank and own calculations

Farms can convert to lower carbon footprint production

The purpose of a tax is to create a financial incentive to change production and thus reduce emissions over time. The more each farm manages to reduce its emissions, the lower the cost of the tax will be. Several actions can help reduce a farm's net emissions. This could include the use of the feed additive Bovaer, which can reduce methane emissions from conventional dairy cows. The effect is included in the results of the analysis.

Reductions can also be achieved by converting production type, for example from animal production to more plant-based production. However, there will be costs associated with this, and there may also be several practical limitations that individual farms will have to deal with. As it is difficult to assess the options for individual farms, this analysis does not take into account the possibility of farms being able to change their line of business.

Planned return of tax proceeds targets farms hit hardest by the carbon tax and struggling to adapt

According to the Agreement on a Green Denmark, the proceeds from the carbon tax in 2030-31 will be returned to investments in such things as climate technology, green initiatives and production conversion targeted at the farms that are hardest hit by the carbon tax and find it difficult to adapt. The return of the proceeds is thus expected to primarily benefit cattle and pig producers, who currently have a high carbon footprint, in a transition period towards more climate-friendly agricultural production. A tax would therefore not take resources out of the industry as a whole, but would involve redistribution between individual farms and operations instead. Possible return of tax proceeds is not included in the results of the analysis. However, all else being equal, it will help improve conditions for the farms that are hit hardest by a tax.

BOX 2

Calculating the economic impact of a carbon tax on farms– how we did it

The introduction of a tax is expected to have a direct effect and several second-order effects that will affect farm finances. The effects are based on several assumptions described in this box.

Pass-through of carbon tax on farm accounts
The starting point for the analysis is farm accounting data for 2021-23. Due to large fluctuations in financial performance from year to year, the calculations are based on an average of the three years. In the projection period from 2025 to 2035, it is assumed that farm accounts are identical year after year. The financial statements, including earnings, equity and liquidity, are only adjusted for the effect of the introduction of the carbon tax on livestock and related consequences. This means that a profit (loss) in a given financial year, besides the carbon tax, does not lead to improved (weakened) equity over time.

Tax rate and base deduction
The Agreement on a Green Denmark includes a carbon tax on agricultural emissions from livestock digestion and manure management. The tax will be phased in gradually from kr. 300 per tonne of CO_2e emitted in 2030 to kr. 750 in 2035. This will be combined with a base deduction covering 60 per cent of the average emissions for the given type of animal. This means that the effective tax rate will be kr. 120 in 2030 and kr. 300 in 2035. In this analysis, the effective tax rate is used in the calculation of the impact on farm accounts.

According to the agreement, the base deduction will limit the increase in costs due to the tax, while maintaining an incentive for individual farms to convert production and use the most climate-efficient operating methods and technologies.

This analysis deals only on the effect of the carbon tax and does not consider the other elements of the political agreement. The Agreement on a Green Denmark also includes the introduction of a subsidy for reduced fertiliser use of kr. 750 per tonne of CO_2e from 2028 and a carbon tax on emissions from agricultural lime of kr. 750 per tonne, which will be phased in gradually from 2028 to 2030.

Emission coefficients for livestock
The carbon tax will cover emissions associated with both livestock digestion and manure management. The design of the tax is simplified to use average emission coefficients for the different types of livestock. The emission coefficients are in accordance with the *Climate Status and Outlook 2022*[1] and coordinated with the Ministry of Food, Agriculture and Fisheries.

Assumptions from GreenREFORM, an environmental and climate economic model for the Danish economy
The introduction of a carbon tax is expected to have several second-order economic consequences for agriculture. The magnitude of these effects comes from GreenREFORM, an environmental and climate-economic model for the Danish economy developed by the DREAM modelling group. The main purpose of the model is to assess the macroeconomic, climate and environmental effects of climate policy proposals. The model was used by the Expert Group for a Green Tax Reform to assess the impact of different tax models related to CO_2e emissions in agriculture.

In GreenREFORM, a separate agricultural module has been developed, consisting of 11 agricultural industries representing different operations. A more detailed description of the model and agricultural modelling can be found on the GreenREFORM website (*link*). In this analysis, the results on an aggregate branch of operation from GreenREFORM are used for all farms within that operation type.

The Green Tripartite Agreement builds on the work of the Expert Group for a Green Tax Reform
In February 2024, the Expert Group for a Green Tax Reform presented three main models for a tax on non-energy-related emissions from agriculture. The tax model adopted politically based on the Green Tripartite Agreement is a version of the models recommended by the expert group. In 2023, Danmarks Nationalbank contributed to the expert group's work by assessing the effects of a carbon tax for Danish banks and mortgage credit institutions.[2]

Change in asset prices of land, buildings and machinery/fixtures
A carbon tax is expected to lead to changes in the value of land, buildings and machinery/fixtures in agriculture. The development in asset prices will lead to changes in the farm's balance sheet in terms of equity and total assets. Asset prices are assumed to develop from the announcement of the tax (i.e. from 2025) until the tax is fully phased in by 2035. Land prices are assumed to grow by around 10 per cent after the announcement, partly supported by forest subsidies (see footnote 15). Land prices are then expected to remain relatively unchanged until 2035. The change in land price is assumed to affect all farms, although the effects will depend on the specific local opportunities, e.g. for afforestation. The agreement on nitrogen emission reductions may also affect land prices, but this effect is not included in the analysis.

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A decrease in asset prices is assumed for buildings and machinery/fixtures respectively. The size of the decrease will depend on the type of farm (cattle, pigs, plants, etc.) and whether the farm is conventional or organic. The assumptions about changes in asset prices are derived from the work of the expert group and model output from GreenREFORM.

Possibility of passing on part of the tax in higher consumer prices
Based on the assumptions in the GreenREFORM model, a tax is also expected to affect both production volumes and prices of agricultural products. A decrease in production would lead to higher prices, and thus, part of the immediate tax burden can partially be passed on to consumers in the form of higher prices. Taken together, these effects will reduce overall earnings in agriculture.

Additives (Bovaer)
The analysis includes the use of the feed additive Bovaer, which has a methane-reducing effect in dairy cows. Research from Aarhus University (AU) concludes that the use of Bovaer reduces methane emissions from dairy cows by 27 per cent.[3] Bovaer is currently not approved for use in organic production.

No farms convert their line of business as a result of a tax
The results of the analysis do not take into account whether farms will respond to a given tax by switching to a less climate-impacting operation.

[1] The Danish Energy Agency, Denmark's Climate Status and Outlook 2022 (*link*).
[2] See Expert Group for a Green Tax Reform – Final report, Feb. 2024, section 7.16 (*link* – only available in Danish).
[3] See e.g. the Danish Veterinary and Food Administration (*link* - only available in Danish) and Aarhus University, DCA National Centre for Food and Agriculture (*link* - only available in Danish).

04

Credit institutions are well equipped to handle losses

With loans totalling kr. 274 billion, upsets in the agricultural sector can potentially affect credit institutions and financial stability. However, the results of this analysis show that institutions are well placed to cover potential losses resulting from the carbon tax on agriculture. This is mainly due to significant collateral/pledges on underlying assets and their ongoing profits.

For institutions, losses could materialise through increased credit risk. If the economy of a farm weakens, the likelihood of it defaulting on its loans will increase. This results in a higher expected loss on loans for the institution, which can increase the institution's vulnerability to other risks.

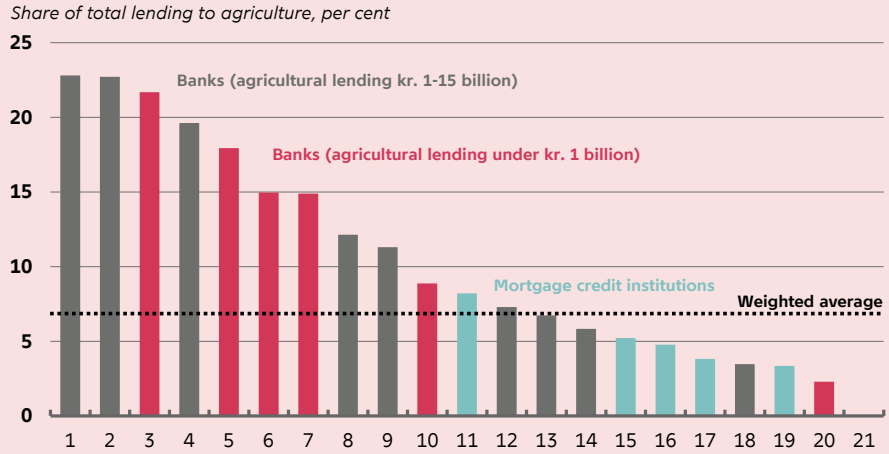
For some banks, insolvent farms will account for more than 20 per cent of agricultural lending

Without lower emissions among farmers, almost 4 per cent of farms will be in risk of insolvency by 2035, see previous chapter. These farms currently have loans totalling around kr. 19 billion corresponding to 7 per cent of the total exposures to agriculture. For some banks, more than 20 per cent of agricultural lending is estimated to be to in risk of insolvency by 2035, see chart 7. For mortgage credit institutions, which account for most agricultural lending, insolvent farms will make up a relatively smaller share.

CHART 7

For some banks, more than 20 per cent of agricultural lending will be to insolvent farms after full phasing in of the tax

Loans to insolvent farms in per cent of total agricultural lending after fully phased in carbon tax in 2035



Note: The institutions are grouped by type of institution and total agricultural lending in kr. billion. The calculations are based on a sample of farms financial accounts, which under certain conditions is assumed to be representative of the individual institution's total agricultural lending. See box 3 for a description of the assumptions behind the calculations.

Source: Danmarks Nationalbank, SEGES Innovation P/S, The expert group for green tax reform, and own calculations

Losses on loans can be covered by impairment charges and collateral

Credit institutions can cover loan losses in several ways, which in this analysis is collectively referred to as their "buffer". Firstly, a credit institution can use already recognised impairment charges. These are funds already reserved to cover loan losses. Secondly, an institution will often have high collateralisation behind the loan in the form of mortgages on the properties being financed (e.g. land and/or buildings). In the event of default, the institution will have the option to liquidate the pledged assets used as collateral. There may be uncertainty about the value of these assets and their actual realisable value.

If the recognised impairment charges and collateral in underlying assets are not sufficient to cover losses, an institution has two additional safeguards available. Initially, the institution can use part of its ongoing profit. If profit is also insufficient, the institution can ultimately make use of its excess capital adequacy. Excess capital adequacy is the excess capital an institution has available over and above the regulatory capital requirement.[15] However, the institution's profits and excess capital adequacy may also be needed for other purposes, such as covering other losses that the institution may face.

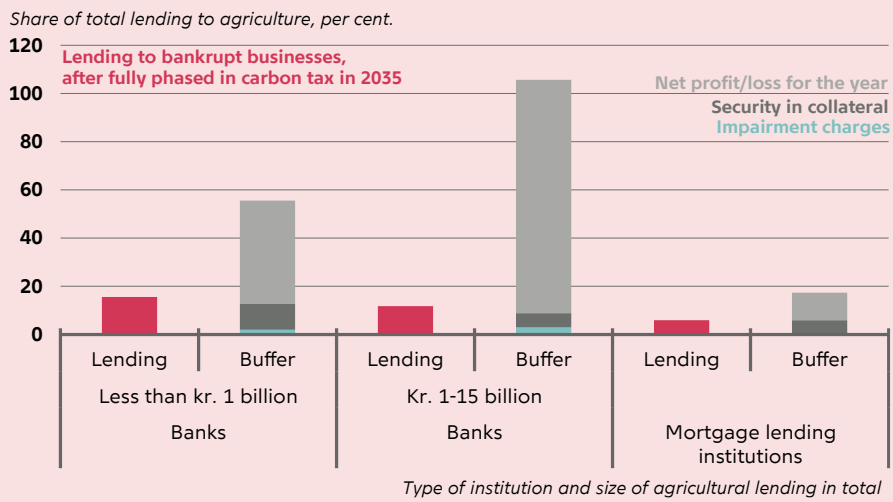
Institutions are generally well equipped to handle losses resulting from a tax

Credit institutions generally have good opportunities to cover losses on loans to agriculture due to a carbon tax, see chart 8. Across institutions, the buffer is over four times larger than potential losses on lending to farms in risk of insolvency.

CHART 8

Credit institutions can handle potential losses on agricultural lending

Lending to insolvent farms after fully phased in carbon tax and funds to cover losses ("buffer")



Note: Institutions are grouped by type of institution and the size of the agricultural loan. For each category, the share of agricultural lending allocated to farms in risk of insolvency after full implementation of the carbon tax is shown. Lending (before impairment charges) is calculated as of the end of 2023 and projected to 2035 based on developments in farm accounts. Institutions are assumed to be able to cover loan losses with already recognised impairment charges, collateral/pledge in underlying assets, and a single year's annual profit. The sum of these is indicated by "buffer" in the chart. Impairment charges, collateral and profits are calculated as a weighted average across the institutions in each category.

Source: SEGES Innovation P/S, the Expert Group for a Green Tax Reform and own calculations.

[15] Credit institutions are subject to requirements from the authorities regarding the size and composition of their capital. See box 3 for more details.

The mortgage credit institutions have lower overall earnings in relation to lending to agriculture than the banks. This is partly because lending business for agriculture constitutes a larger part of the income-generating activity in mortgage credit institutions than is the case in banks.

Potential losses can take a toll on the resilience of individual banks

For some banks, a high exposure to agricultural customers can mean that the potential losses can be so large that the institution's impairment charges and collateral are insufficient. In such a situation, it may be necessary for the institution to use profits and potential excess capital adequacy relative to applicable capital requirements. In cases where losses on agricultural customers significantly erode an institution's profits and part of its excess capital adequacy, it can weaken resilience in the event of a simultaneous or subsequent economic downturn.

Mortgage credit institutions are well placed to handle potential losses

Impairment charges and the collateral behind the loan will be sufficient to cover losses for mortgage credit institutions. At the same time, three conditions support the mortgage credit institutions' possibility to cover losses. Firstly, the maximum loan limit in Denmark for mortgage credit institutions towards agriculture is 70 per cent in relation to the value of the property. Secondly, they also often hold primary collateral in the priority ranking, which means that the institution has the first right to use pledged assets to cover losses if the borrower defaults. For the same reason, mortgage credit institutions often have less funds set aside for future impairment charges than banks.

Finally, a group-affiliated or cooperating bank will also often be the intermediary for the mortgage loan. As part of the brokerage agreement, the bank has provided a certain loss guarantee or similar behind the mortgage loan, which will make the bank obligated to bear part of the mortgage credit institution's losses.

Results of the analysis may overestimate the impact of the carbon tax on agriculture

The results of the analysis may overestimate the effects of a tax. Firstly, the outset for the calculations are that credit institutions may incur losses on *all* lending to farms in risk of insolvency. However, being in risk of insolvency does not mean that an individual farm will actually go bankrupt. Secondly, the results imply a conservative assumption that the carbon tax does not lead to a transition in agriculture and thus fewer emissions, which is the overall purpose of a tax. Fewer emissions will mean less cost associated with a tax. Finally, a unchanged loan portfolio and buffer are assumed for the institutions. The three assumptions contribute to the results indicating an upper bar for the effect of the tax introduced. See more in box 3.

Gradual phasing in of the tax increases the institutions' room to manoeuvre

Potential credit losses for the institutions will not materialise the moment a carbon tax is announced or phased in. The projection shows that credit losses are likely to be spread over a number of years.

The announcement and the gradual phasing in period of the tax thus supports financial stability. This allows banks and mortgage credit institutions to absorb potential losses over several years. The announcement of the tax and the gradual phasing in will also give institutions more time to adjust their loan portfolios and capital planning, and agriculture will have more time to transition production in a greener and less carbon-intensive direction.

BOX 3

Calculating potential loan losses for institutions – how we did it

Linking farm data with the credit register
Loans to farms in risk of insolvency for the individual bank or mortgage credit institution are calculated by linking calculations at farm level with information about farm debt etc. in the credit register.[1] Loans are calculated at the end of 2023.

The proportion of farms in risk of insolvency is calculated based on the sample of farm accounts from SEGES Innovation P/S. The calculation is based on data such as the number of animals on each farm and is calculated at institution level, broken down by farm type. Since the results are based on a sample of farms, which is then used as a proxy for the institution's entire agricultural lending, there are some uncertainties associated with the results. The actual exposure to farms in risk of insolvency for each institution may differ from the sample prediction. A number of adjustments have been made as a result of this, taking into account, among other things, the distribution of impairment stages on the institution's entire lending to agriculture within the individual farm types. For some institutions, the sample size is considered too small and results are therefore not shown. The calculations only include direct lending to agriculture by credit institutions. Loans to related sectors are disregarded.

Lending to farms in risk of insolvency
The analysis assumes that credit institutions incur losses on *all* lending to farms in risk of insolvency. This implies an assumption that all farms in risk of insolvency will default on their debt, i.e. the loan is given a probability of default (PD) of 100 per cent. This means that the total outstanding debt is written down less the institution's collateral in underlying assets. However, being in risk of insolvency does not mean that an individual farm will go bankrupt, but it is a strong indication that the farm is under financial pressure and that the current business model may be unsustainable. Intermediation agreements between banks and mortgage credit institutions, including loss guarantees etc., are not taken into account. Mortgage credit institutions are therefore assumed to bear the full risk of the total mortgage loan, and not the bank that brokered the loan to the customer.

Impairment charges
The institutions' already recognised impairment charges on their agricultural customers are calculated based on the accumulated impairments for the individual loans reported to the credit register. Any management surcharges/estimates related to the institutions' agricultural lending are taken into account.

Collateral securing against credit losses
When banks and mortgage credit institutions grant loans to companies, for example, they typically require collateral for their loans in the form of mortgages on underlying assets. This could be collateral/pledges in land, buildings, machinery, etc. In this analysis, the value of an institution's collateral is calculated by the part of a loan's outstanding debt in which the institution has allocated collateral/pledge. The value of collateral for a given loan cannot be used to cover losses on another loan, and the value of collateral that exceeds the value of the outstanding debt on the individual loan is not included in the analysis.

In the event of default, the institution will have the option to liquidate the pledged assets used as collateral. There may be uncertainty about the value of these assets and their actual realisation value. The value will depend on general market conditions, among other things. The value of the institutions' collateral is based on the current level and is not adjusted based on the carbon tax.

Profit
The analysis uses an average of the institutions' profit before tax for the three financial years 2021-23.

Excess capital adequacy relative to applicable capital requirements
Credit institutions are subject to a number of requirements from the authorities regarding both the size and composition of their capital. Capital requirements are designed to ensure that institutions have sufficient capital for the risks related to their exposures. Their capital requirements are determined based on an individual solvency need and a number of capital buffer requirements, including a capital conservation buffer, a countercyclical capital buffer and an SIFI buffer. Excess cover is calculated as the total capital requirement, including capital buffers. The risk-weighted exposures (RVE) are assumed to remain unchanged in the projection.

[1] The credit register (*link*) is a report to the Danish Financial Supervisory Authority and Danmarks Nationalbank that contains detailed information on monetary financial institutions' loans and credits to all types of debtors and for all types of loans/credits.

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